

Security implementation

Security Overview

This document describes the permissions and roles assigned to protect the foodhub database against unauthorised access and ensure that only authorised roles can access and modify sensitive data.

The users who have access to the database are:

Staff: Partial access. Staff users are allowed to INSERT, UPDATE, and SELECT records, but they are not allowed to Delete data. They have access to the main operational tables but not to catalogue tables. For catalogue tables, only SELECT operations are permitted.

Manager: Full access to non-technical tables. Managers can also INSERT, UPDATE, and DELETE records in catalogue tables in order to manage and update reference data.

Login Creation

Database logins were created at the SQL Server level to authenticate authorized users before granting access to the FoodHub database.

Separate credentials were configured for:

- Management personnel
- Operational staff

Each login includes:

- Unique username
- Secure password authentication
- Restricted server access

Manager login:

Login - New

Select a page

General

Server Roles

User Mapping

Securables

Status

Connection

Server:
MSI

Connection:
MSI\luisa

[View connection properties](#)

Progress

Ready

Script

Help

Login name:

ManagerFoodHub

Search...

☐ Windows authentication

☐ Microsoft Entra authentication

☒ SQL authentication

Password:

.....

Confirm password:

.....

☐ Specify old password

Old password:

☒ Enforce password policy

☒ Enforce password expiration

☒ User must change password at next login

☐ Mapped to certificate

☐ Mapped to asymmetric key

☐ Map to Credential

Mapped Credentials

Credential

Provi

Add

Default database:

Foodhub

Default language:

<default>

Remove

OK

Cancel

Staff login:

Login - New

Select a page

General

Server Roles

User Mapping

Securables

Status

Connection

Server:
MSI

Connection:
MSI\luisa

[View connection properties](#)

Progress

Ready

Script

Help

Login name:

StaffFoodHub

Search...

☐ Windows authentication

☐ Microsoft Entra authentication

☒ SQL authentication

Password:

.....

Confirm password:

.....

☐ Specify old password

Old password:

☒ Enforce password policy

☒ Enforce password expiration

☒ User must change password at next login

☐ Mapped to certificate

☐ Mapped to asymmetric key

☐ Map to Credential

Mapped Credentials

Credential

Provi

Add

Default database:

Foodhub

Default language:

<default>

Remove

OK

Cancel

User creation

Database users were configured

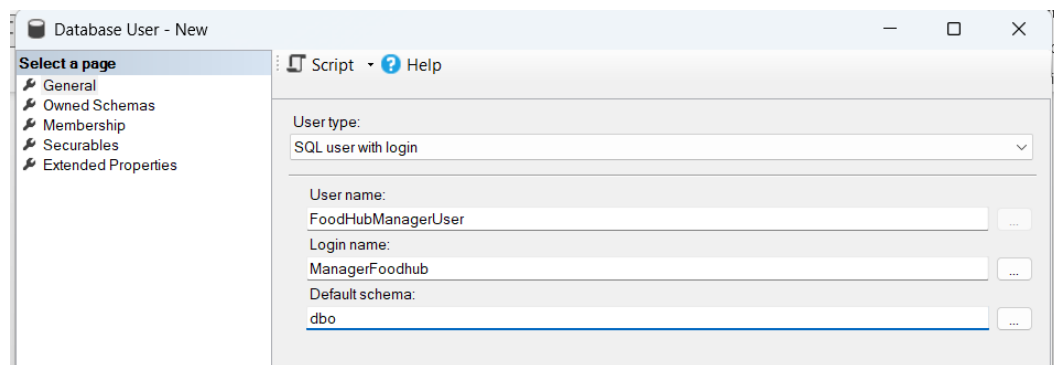
Each database user was linked to its corresponding SQL Server login to establish authenticated database access.

This separation between server authentication and database authorization improves security management and allows permissions to be controlled more efficiently.

Configured Users

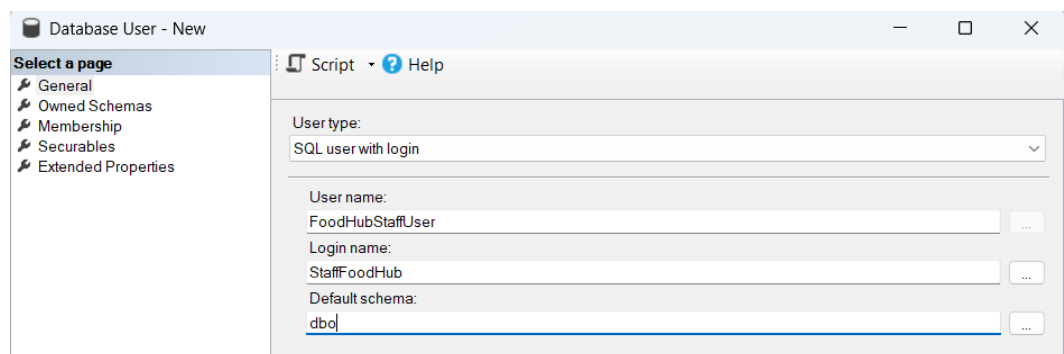
- FoodHubManagerUser
- StaffFoodHubUser

Manager User



The screenshot shows the 'Database User - New' dialog box in SQL Server Enterprise Manager. The 'General' tab is selected in the left-hand pane. The 'User type' is set to 'SQL user with login'. The 'User name' field contains 'FoodHubManagerUser'. The 'Login name' field contains 'ManagerFoodhub'. The 'Default schema' field contains 'dbo'.

Staff User



The screenshot shows the 'Database User - New' dialog box in SQL Server Enterprise Manager. The 'General' tab is selected in the left-hand pane. The 'User type' is set to 'SQL user with login'. The 'User name' field contains 'FoodHubStaffUser'. The 'Login name' field contains 'StaffFoodHub'. The 'Default schema' field contains 'dbo'.

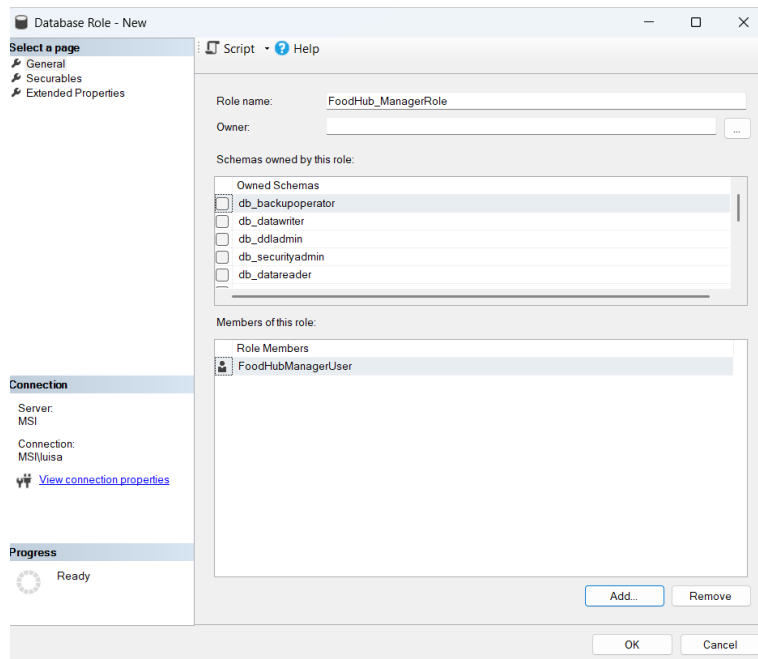
Role-Based Security Configuration

To simplify permission management and improve maintainability, database roles were implemented.

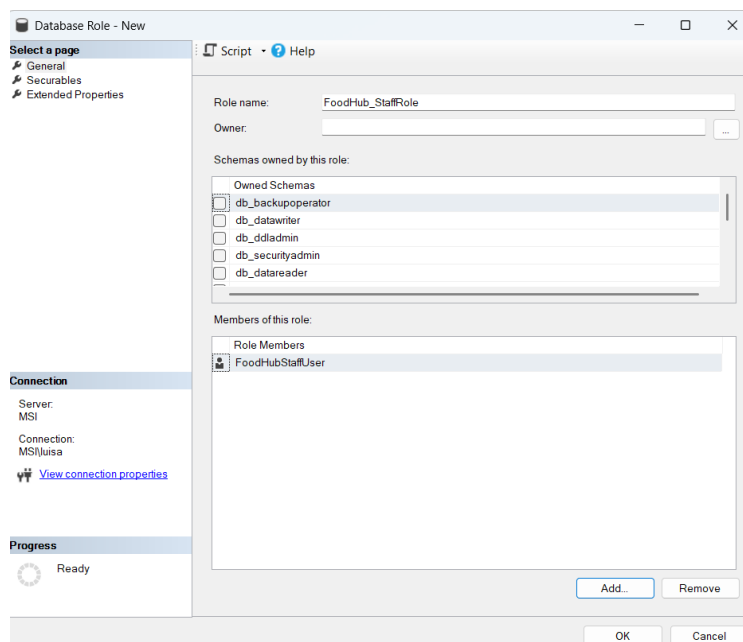
permissions were grouped according to operational responsibilities.

The following custom roles were created:

Manager Role:



Staff Role:



Permission Management

Permissions were assigned according to the principle of least privilege, ensuring users only receive the access required for their responsibilities.

Manager Permissions

Managers were granted SELECT, INSERT, UPDATE, and limited DELETE permissions on operational and catalogue tables.

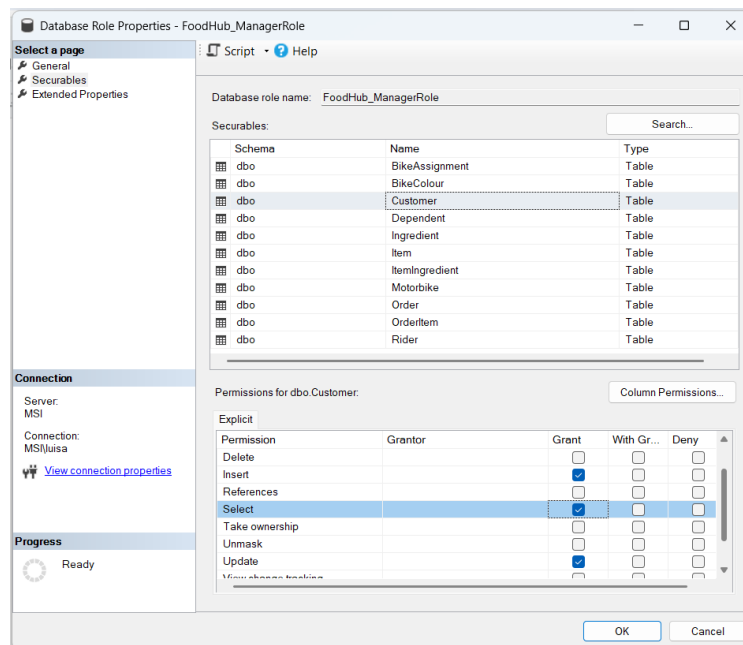
Managers also received access to catalogue and reference tables in order to maintain operational data consistency.

Staff Permissions

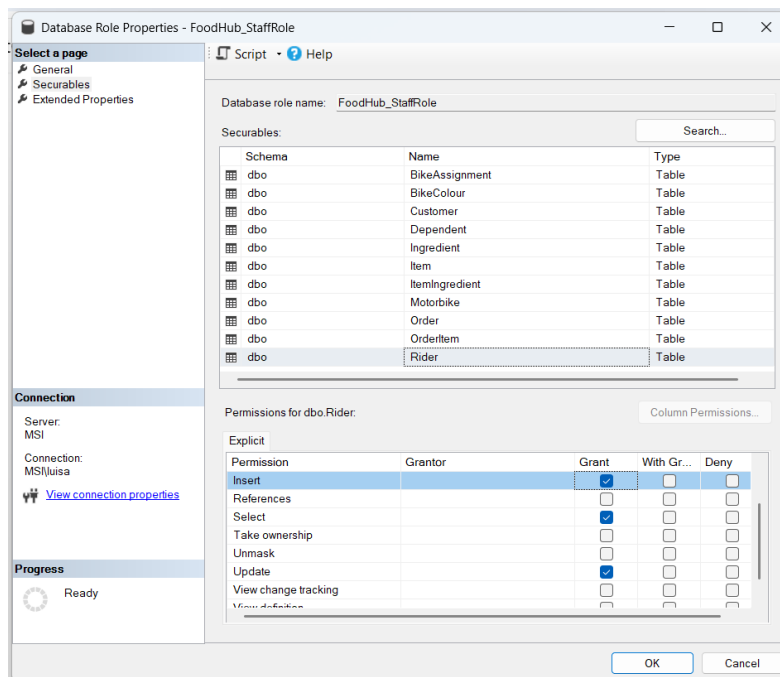
Staff users were granted SELECT, INSERT, and UPDATE permissions on operational tables, while DELETE and catalogue modification permissions were restricted.

This permission structure strengthens database integrity while maintaining operational efficiency.

Manager Role Permissions:



Staff Role Permissions:



4.6 Security Evaluation

The implemented security model significantly improves the protection and reliability of the FoodHub database system.

The use of:

- authenticated logins,
- database users,
- role-based access control,
- and permission restrictions

ensures that system access is controlled according to organizational responsibilities.

Overall, the implemented security mechanisms provide:

- improved access control,
- stronger data protection,
- enhanced accountability,
- and better operational security management.